



Types of Dashboards Transcript

Hello everyone, welcome to BITS Pilani digital. Today we will continue our learning journey for the course data visualisation and storytelling. We will be now going into a module called building dashboards.

In this module, we will be trying to understand what are the different types of dashboards, how dashboard building is a unique way of putting across information to the audience and helping them to drive insights. If you recall from our earlier classes, we have seen about two types of narratives in storytelling. In data storytelling, there are two narratives that are often used.

One is author driven, another is reader driven. However, dashboarding I would say is mixture of both. You have an author driven template, but giving the flexibility of for the reader to make his own narrative out of the data.

This is enabled by creating interactivity by posing suitable elements on to the dashboard design, so that the author and the reader both have common purpose in reading the data. We will now look into what are the types of dashboard and what are the characteristics of the dashboard. First of all, what is a dashboard? You can think dashboard as a collection of small groups of visualisations and a couple of graphs, maybe graphs together with some text or some pictorial presentation along with some narration.

Effectively, these together are trying to tell across a story or enable some insights to the audience. However, what makes dashboards unique is their interactivity. So, when we speak about dashboards, we are talking about interfacing very important information in a limited space and a limited time.

It is very important that the dashboards are very attention seeking in the sense they grab your attention quickly and point you to the insights they want you to look at. In this way, the key features of a dashboard design or a dashboard are rapid scanning, real-time visibility and action orientated design. The best dashboards are concise, data rich and are easily scannable.

Now, let us look at 5 key properties of a dashboard or what makes a dashboard really effective. They are typically small, concise and they are very clear, they are very intuitive and customised. Now, what I would like to take a pause here and connect with the principles of effective visual design, various pre-attentive attributes that we have discussed, the storytelling framework, the different types of narratives and the need to communicate effectively.

All these elements come together on a dashboard. Dashboard is just not presentation of a group of select visuals, but binding them together in a particular context and in a

seamless fashion along with interactivity trying to make data move into insight and driving action from there on. So, keep all these aspects in mind when you are looking at any dashboard.

Again, I am repeating a effective dashboard is small, not in terms of the area that it covers, but in terms of the content that it presents. It is very clear and concise and it has an intuitive appeal and it is customised. When you look at a dashboard, you should intuitively get the logic or the narrative that the author is trying to get across.

Look at this example on the screen. So, this is a dashboard that talks about CO2 emissions per country on any given industrial sector. This is sourced from the IMF, International Monetary Fund's data section.

So, this dashboard simply has 4 visuals 1, 2, 3, 4. The first visual is trying to capture CO2 emissions multipliers for selected country and industry. Then you have CO2 emission intensities and CO2 emissions. So, these are all CO2 emissions varied by definition and type and then you have a comparison of the country with the top 10 emitters in that particular segment.

How do you choose? There is an interactive element wherein the audience or the viewer, the reader in this case can choose the country and also the industry subtype and the year. So, it is showing primarily a comparison graphs that is why you have bar plots and changes over time which is a line graph and it is using all the visual effective visualisation principles and also following the close order from the gestalt principles of visual encoding. There is clear contrast in the graphs.

They are ordered in a logical fashion especially the top 10 countries and what could have been more appealing is had a country that has been selected could have been highlighted in a different colour on the visual here it would have been even more appealing. So, like this when you look at a dashboard you can look at the key elements. It is small in the sense it is fitting all the information in one page.

It has 4 clear concise and clear attributes or metrics that it is talking about and these metrics are not conflicting or dominating each other. They are evenly balanced and the communication is very clear and it has interactivity also. So, when we look at a dashboard that is very simple you can intuitively compare okay I have selected India and I have selected air transport industry.

I know the intensity has been falling over a period of time and India is among the top 4 contributors to CO2 emissions as far as air transport industry is concerned, but it is much lesser than the Ireland. Now intuitively they should question and they should raise a question why Ireland has the top 10, why Ireland is emerging as a top 10 emitter of CO2 when you compare CO2 by plane type or travel. Ireland is a small country located in the northern European region and how come it is the top contributor to CO2.

This kind of inquisitive question should come to your mind when you look at a dashboard. Here there is one more scratch because some of the airlines have their lease business from Ireland because of tax reasons and this might be one of the reasons why you see Ireland having top CO2 emissions because many of the aircraft companies are located or have corporate office in Ireland and that is the reason why you can see CO2 emissions in Ireland are very high as compared to India though the travel numbers would definitely be very higher in India or China. Now let us look at why different types of dashboards exist.

Now look at the fundamental design principle that was used for understanding effective visual design wherein it has been clearly specified that form should follow function which means the form in which you are designing your visual should be determined by the function it is expected to deliver. Put it in other words function drives the form. In case of dashboards also you should ask the fundamental question like in any data visualisation exercise which is supposed to be audience centric what is the function that you want to achieve from the dashboard.

Once you answer that question you can then build up a dashboard or get a context around the dashboard to what kind of visuals you have to put in and how to interact them and connect them into a good story. If you are talking to top management of a company they might be interested in the looking at the overall picture comparison with the competitors or with the regulation compliance so their outlook would be broader strategic forward looking and they have details at the aggregate level without going into the micro details. As opposed to this if you have an internal analyst who is looking at the operations he would be more interested in to understand how sales is growing up revenue targets are met or what are the reasons for different kind of search engine actions how the website is traffic is coming in which regions of the website are active how to optimise them.

This kind of analysis a dashboard that captures these things would be little different where the analyst is looking trying to create a correlation or to establish patterns and see how the over a period of time they are changing and bring insights to the decisioning board to optimise the operations. So this kind of a dashboard would be different from that where you would show to a top management the top management person would look at an aggregate data the analyst would look at the aggregate data but then drill down to a level of sophistication where he can give clear operational insights. As opposed to this you have operations dashboard which are more concerned with keeping up of things or running day-to-day businesses they are not worried about creating strategic leaps but are looking at meeting the compliance parameters or ensuring the uptime of the website.

Those dashboards would be more real-time and tuned to tracking minute details where if a deviation they would flag and correct it so the performance can be kept. So when



you see the dashboard is a common element across all these three types but the elements that would go into these three type of dashboards are different because the functions it is expected to deliver are entirely different. So we have to see what is the key factor that determines the dashboard type.

First is the time sensitivity of the analysis. For example a strategy requires a time frame which is longer as compared to an operations time frame which is mostly real time or daily or even for a matter of hours. Then you have level of interactivity the strategy dashboard would not require high level of interactivity in some sense it would be more of an aggregate phenomenon looking at combined interactivity at the regional level maybe at the department level.

And nature of decision supported strategy would be more forward looking it would typically follow follow-up exercise by the analyst or the executive team to answer certain questions that the top management is raising. As opposed to that the analyst and the operating dashboard or operators using the dashboard would have very high time sensitivity and the nature of decisions would be very critical in the sense they have specified decisions to be taken if the data or the dashboard points to something.

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